

HDOC Day-31

Write a C program to perform the following menu-driven operations on a given 2D array:

1. Find the sum of all elements in a matrix.
2. Find the sum of each row of a matrix and store it in an array.
3. Find the transpose of a matrix.
4. Exit

5. **Sample Output:**

6. Enter the number of rows and columns: 2 3
7. Enter the matrix elements:
8. 1 2 3
9. 4 5 6
10. Enter your choice: 1
Sum of all elements = 21
11. Enter your choice: 2
12. Row sums:
13. Sum of row 1 = 6
14. Sum of row 2 = 15
15. The row-sum array contains: [6, 15]
16. Enter your choice: 3
17. Transpose of the matrix:
18. 1 4
19. 2 5
20. 3 6

21.2. Algorithm

Step 1: Start

Declare a 2D array `matrix`, a row-sum array, and variables for rows, columns and calculations.

Step 2: Input

Read the number of rows and columns and enter the matrix elements.

Step 3: Display Menu

1. Sum of all elements
2. Sum of each row
3. Transpose of matrix

4. Exit

Step 4: Perform Operation

Choice 1:

Traverse every element of the matrix and add them to sum. Display the total.

Choice 2:

For each row, calculate its sum and store it in rowSum[]. Display the row sums.

Choice 3:

For every element matrix[i][j], print it as matrix[j][i] to obtain the transpose.

Choice 4:

Terminate the program.

Step 5:

If the choice is invalid, display an appropriate message.

Step 6:

Stop.

● C Program:

- #include <stdio.h>
-
- int main()
- {
- int matrix[10][10], rowSum[10];
- int rows, cols;
- int i, j, choice;
- int sum;
-
- // Input number of rows and columns
- printf("Enter the number of rows and columns: ");
- scanf("%d %d", &rows, &cols);
-
- // Input matrix elements
- printf("Enter the matrix elements:\n");
- for (i = 0; i < rows; i++)
- {
- for (j = 0; j < cols; j++)

```

•     {
•         scanf("%d", &matrix[i][j]);
•     }
• }
•
• // Menu-driven program
• do
• {
•     printf("\n----- MENU ----- \n");
•     printf("1. Sum of all elements\n");
•     printf("2. Sum of each row\n");
•     printf("3. Transpose of matrix\n");
•     printf("4. Exit\n");
•
•     printf("Enter your choice: ");
•     scanf("%d", &choice);
•
•     switch (choice)
•     {
•         case 1:
•             // Calculate sum of all elements
•             sum = 0;
•
•             for (i = 0; i < rows; i++)
•             {
•                 for (j = 0; j < cols; j++)
•                 {
•                     sum = sum + matrix[i][j];
•                 }
•             }
•
•             printf("Sum of all elements = %d\n", sum);
•             break;
•
•         case 2:
•             // Calculate sum of each row
•             printf("Row sums:\n");
•
•             for (i = 0; i < rows; i++)

```

- {
- rowSum[i] = 0;
-
- for (j = 0; j < cols; j++)
- {
- rowSum[i] = rowSum[i] + matrix[i][j];
- }
-
- printf("Sum of row %d = %d\n", i + 1, rowSum[i]);
- }
-
- printf("The row-sum array contains: [");
-
- for (i = 0; i < rows; i++)
- {
- printf("%d", rowSum[i]);
-
- if (i < rows - 1)
- printf(", ");
- }
-
- printf("]\n");
- break;
-
- case 3:
- // Display transpose
- printf("Transpose of the matrix:\n");
-
- for (i = 0; i < cols; i++)
- {
- for (j = 0; j < rows; j++)
- {
- printf("%d ", matrix[j][i]);
- }
- printf("\n");
- }
-
- break;
-

- case 4:
- printf("Exiting the program...\n");
- break;
-
- default:
- printf("Invalid choice! Please enter a choice from 1 to 4.\n");
- }
-
- }while (choice != 4);
-
- return 0;
- }

Test Case Table:

Test Case	Input rows & columns	Matrix	Choice	Expected Output
1	2 3	1 2 3 4 5 6	1	Sum of all elements = 21
2	2 3	1 2 3 4 5 6	2	Row sums = [6, 15]
3	2 3	1 2 3 4 5 6	3	Transpose = 1 4 2 5 3 6
4	2 3	1 2 3 4 5 6	4	Program exits
5	2 3	1 2 3 4 5 6	5	Invalid choice message

Compilation and Execution:

Test Case 1:

```
File System      demo1
(kali㉿kali)-[~]
└─$ ./a.out
Enter the number of rows and columns: 2 3
Enter the matrix elements:
1 2 3 4 5 6

——— MENU ———
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 1
Sum of all elements = 21
```

Test Case 2:

```
——— MENU ———
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 2
Row sums:
Sum of row 1 = 6
Sum of row 2 = 15
The row-sum array contains: [6, 15]
```

Test Case 3:

```
——— MENU ———
1. Sum of all elements
2. Sum of each row
3. Transpose of matrix
4. Exit
Enter your choice: 3
Transpose of the matrix:
1 4
2 5
3 6
```

Test Case 4:

```
—— MENU ——  
1. Sum of all elements  
2. Sum of each row  
3. Transpose of matrix  
4. Exit  
Enter your choice: 4  
Exiting the program...
```

Test Case 5:

```
(kali@kali)-[~]  
$ ./a.out  
Enter the number of rows and columns: 2 3  
Enter the matrix elements:  
1 2 3 4 5 6  
  
—— MENU ——  
1. Sum of all elements  
2. Sum of each row  
3. Transpose of matrix  
4. Exit  
Enter your choice: 5  
Invalid choice! Please enter a choice from 1 to 4.
```